

Smart Bus Transport System

Project Analysis Report

1. Overview

The Smart Bus Transport System is a PHP-based transport booking and management application designed to support passenger booking, route management, schedule operations, bus monitoring, and payment tracking. The project combines a web interface with a MySQL database to manage daily transport operations in a simple but functional way.

The application is organized around a central ticketing and scheduling workflow. Admin and operator users manage bus inventory, routes, and schedules, while passengers search available trips and book seats online.

2. Project Objectives

- manage public transport routes and schedules;
- keep bus and driver records in one place;
- enable passengers to search trips and reserve seats;
- record bookings, payments, and tickets;
- improve operational visibility for transport staff.

3. Current System Architecture

The project uses a lightweight web architecture:

- Frontend: PHP pages rendered directly in the browser
- Backend: PHP server-side logic and database queries
- Database: MySQL using SQL schema scripts
- Session handling: PHP sessions for login and role-based access
- Security: CSRF tokens, password hashing, and validation checks

The structure is clear, with core files such as `index.php`, `dashboard.php`, `passenger-dashboard.php`, `includes/db_connect.php`, `config/config.php`, and `database/smart_bus.sql`.

4. Users and Roles

- **Admin:** manages overall system operations, bus and route records, and operational data.
- **Operator:** supports scheduling and service tracking.
- **Passenger:** searches trips, books seats, and receives tickets.

5. Functional Module Analysis

Authentication and Access Control

The login function validates email and password against the users table. Passwords are hashed using PHP `password_hash` and `password_verify`, which is a strong basic practice. The session stores `user_id`, `user_name`, and `user_role`, and the application redirects based on role.

Bus Management

Bus data includes registration number, make, model, year, bus type, seating capacity, and status. Admin users can register buses and monitor their availability. This supports fleet management and helps operators know whether a bus is available, assigned, or inactive.

Route Management

Routes are defined with codes, origins, destinations, districts, distance, estimated duration, and fare. This enables trip planning and supports route-based ticketing.

Schedule Allocation

Schedules link a route, bus, travel date, departure time, and arrival time. Available seats are stored and deducted as bookings are completed.

Booking and Payment

Passengers can search scheduled trips based on origin, destination, district, date, and bus. Booking logic checks for seat availability, valid seat numbers, and duplicate bookings. Payment is processed at the same time as booking creation, and a ticket record is generated.

Ticket Generation and Travel History

Tickets are created for each successful booking. Travel history stores completed journeys for the passenger and supports review of prior trips.

6. Database Design Review

The database schema includes the main operational tables: users, drivers, buses, routes, schedules, bookings, payments, tickets, and `travel_history`.

The design is logically organized and covers the full lifecycle of a trip from route creation to completed travel. Key relational links are present through foreign keys, and uniqueness constraints are used to prevent duplicate records such as repeated seat bookings on the same schedule.

The schema is suitable for a small to medium-sized public transport system and is simple enough for a student project or early business prototype.

7. Strengths of the System

- Clear separation of admin and passenger workflows
- Role-based redirect logic after login
- CSRF protection on form submissions
- Password hashing for secure credentials
- Functional booking and payment flow
- Realistic database schema with proper constraints
- Configurable environment-based database setup

8. Weaknesses and Gaps

- Code duplication across files
- Limited modular structure
- Insufficient production-level security
- Administrative features are still basic
- Payment system is simulated rather than integrated
- Limited reporting and analytics
- Need for better mobile and responsive design

9. Recommendations

- adopt a cleaner MVC or service-oriented structure;
- centralize validation and database access logic;
- add audit trails and admin activity logging;
- integrate real payment channels and notification services;
- add route analytics, occupancy reports, and driver assignment dashboards;
- improve responsiveness and accessibility for mobile users;
- add automated tests for booking logic and authentication flows.

10. Conclusion

The Smart Bus Transport System demonstrates a strong foundation for a public transport booking and scheduling platform. It has clear functional goals, a practical database design, and a usable user flow for both passengers and administrators.

The project is especially suitable as a prototype or academic system, and with additional improvements in architecture, reporting, and payment integration, it can evolve into a more robust real-world transport management solution.

11. Overall Assessment

The application is effective as a functional prototype, with solid basics in authentication, route and bus management, booking, and ticketing. It is well organized for a small project but would benefit from stronger modular design and production readiness measures before commercial deployment.